Decision and Risk Analysis

Review of Evaluating Regression Models

And More Practice Questions

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Before trusting a Bayesian regression model, ask **three questions**:

1. **How fair is it?**
   Data collection, purpose, who is affected, embedded biases.
2. **How wrong is it?**
   Are the assumptions reasonable? $\Rightarrow$ posterior predictive checks.
3. **How accurate are its predictions?**
   Typical error and interval coverage $\Rightarrow$ cross-validation, ELPD.

"All models are wrong, but some are useful."

Running example: Capital Bikeshare

- 500 days (2011–2012);
  $Y =$ daily rides, $X =$ feels-like temperature.
- Normal regression:

$$Y_i \mid \beta_0, \beta_1, \sigma \sim N(\mu_i, \sigma^2), \quad \mu_i = \beta_0 + \beta_1 X_i$$

- Priors:

$$\beta_{0c} \sim N(5000, 1000^2),\ \beta_1 \sim N(100, 40^2),\ \sigma \sim \text{Exp}(0.0008)$$

- 20,000 posterior draws of $(\beta_0, \beta_1, \sigma)$ via MCMC.

Q1. Is the model fair?

- How was the data collected? By whom, and for what purpose?
- How might results impact individuals and society?
- What biases might be baked into the data or the analysis?

Bikes: no obvious ethical issue, BUT the data only reflects people who already use the service.

Examples of unfair models: résumé screening favoring men, face recognition failing on underrepresented groups, a detention algorithm recommending detention almost always.

Q2. How wrong is the model? Assumptions

The Normal regression model assumes:

1. **Independence:** Y_i independent given X_i.
2. **Linearity:** $E(Y \mid X) = \beta_0 + \beta_1 X$.
3. **Normality & constant variance:** Y varies normally around μ with the same σ for all X.

Scatterplots help, but a Bayesian has a better tool ...

Posterior predictive checks (PPC)

Idea: if the model is reasonable, data it simulates should look like the data we observed.

1. For each posterior draw $(\beta_0^{(j)}, \beta_1^{(j)}, \sigma^{(j)})$, simulate $Y_i^{(j)} \sim N(\beta_0^{(j)} + \beta_1^{(j)} X_i, (\sigma^{(j)})^2)$ for every observed X_i.
2. Overlay the simulated datasets on the observed data (`pp_check()`).

Bikes: simulations capture center and spread, but miss the *bimodality* of ridership. Not perfect ... still useful.

PPC: what to look for

- **Shape:** skew, multiple modes, heavy tails.
- **Summary statistics** $T(y)$: compare observed mean, SD, min, max to their simulated distributions (`pp_check(fit, plotfun = "stat", stat = "sd")`).
- **Impossible values:** e.g. negative ride counts $\Rightarrow$ wrong data structure.
- **Patterns vs.** X**:** does spread grow with X? (non-constant variance).

Caveats:

- The same data both fit and check the model $\Rightarrow$ PPC is lenient; passing $\neq$ "true" model.
- Ask: are the mismatches large enough to matter *for our goal*?

When the model is (too) wrong

- **Dependence** (grouped, temporal, spatial)
  $\Rightarrow$ hierarchical, time-series, or spatial models.
- **Wrong data structure** (counts, binary)
  $\Rightarrow$ Poisson, Negative Binomial, logistic (Ch. 12–13).
- **Nonlinearity / non-constant variance / skew**
  $\Rightarrow$ transform Y, X, or both (e.g. $\log Y$).

Q3. Accuracy for a single day

Oct 22, 2012: 75.46°F, observed $Y = 6{,}228$ rides.

- Posterior predictive mean $Y' = 3{,}967$
  $\Rightarrow$ error $Y - Y' = 2{,}261$ rides.
- Posterior predictive SD $= 1{,}280$
  $\Rightarrow$ scaled error $= 2261/1280 \approx 1.77$ SDs.
- 50% interval $(3096, 4831)$: Y **outside**.
  95% interval $(1500, 6482)$: Y **inside**.

Key point: raw error alone is misleading. The same 2,261 error with SD 500 would be 4.5 SDs off. *Scale matters.*

Accuracy across all data: summaries

Metric	Meaning	Bikes
MAE	median $\lvert Y_i - Y_i' \rvert$	989.7
Scaled MAE	median $\lvert Y_i - Y_i' \rvert / \text{sd}_i$	0.771
within_50	share inside 50% interval	0.438
within_95	share inside 95% interval	0.968

- Ideal coverage: within_50 ≈ 0.5, within_95 ≈ 0.95.
- Computed with `prediction_summary()`.

Cross-validation (honest evaluation)

Evaluating on the data used to fit is **optimistic**.

k-**fold CV:**

1. Split data into k folds.
2. Train on $k-1$ folds, test on the held-out fold.
3. Repeat so each fold is tested once; average metrics.

	In-sample	10-fold CV
MAE	990	1029
Scaled MAE	0.771	0.802
within_50	0.438	0.420
within_95	0.968	0.968

Choosing k

- **Small k:** train on less data $\Rightarrow$ pessimistic (biased) error; cheap.
- **Large k (LOO, $k = n$):** low bias, but n refits and noisier estimates.
- **Default:** $k = 10$ (or 5 if fitting is slow); larger k / LOO for small n.
- **Bayesian shortcut:** `loo()` approximates LOO from one fit. `kfold()`.
- **Fold structure beats k:** time series $\rightarrow$ forward-chaining; grouped data $\rightarrow$ leave groups out.

ELPD

MAE judges only a point prediction. ELPD judges the *whole* predictive distribution.

Posterior predictive density of a new value y' (averages over parameter uncertainty):

$$f(y' \mid \vec{y}) = \iiint f(y' \mid \beta_0, \beta_1, \sigma)\, f(\beta_0, \beta_1, \sigma \mid \vec{y})\, d\beta_0\, d\beta_1\, d\sigma$$

- Ask: how much probability density did the model put *where the new value actually landed*?
- Rewards being accurate **and** honestly uncertain:
  - off-center prediction $\Rightarrow$ low density;
  - overconfident (too narrow) $\Rightarrow$ very low density when it misses;
  - too vague (too wide) $\Rightarrow$ low density everywhere.

Computing and using ELPD

We don't have new data, so use leave-one-out:

$$\widehat{\text{ELPD}} = \sum_{i=1}^{n} \log f(y_i \mid \vec{y}_{-i})$$

- Each y_i is scored by a model fit *without* it; `loo()` approximates this from one fit (PSIS).
- **Higher (less negative) is better.**
- Bikes: ELPD $= -4289$ (SE 13.1), about -8.6 per day. LOOIC $= -2 \times$ ELPD $= 8578$.
- No absolute scale: the number depends on n and units of Y $\Rightarrow$ use it to *compare* models on the same data.
- Compare with `loo_compare()`: a difference smaller than about 2 SE of the difference is not convincing (Ch. 11).

Improving accuracy & two separate questions

Improve predictions:

- More data (sharpens a real relationship; can't create one).
- Better / more predictors (e.g. humidity) ... up to a point.

Don't confuse:

- *MCMC quality:* is the simulation a good approximation of the posterior? (trace plots, $\hat{R}$, ESS)
- *Model quality:* is the model fair, reasonable, accurate?

A great approximation of a bad model is still a bad model.

Recap

1. **Fair?** Context-dependent: data, purpose, consequences.
2. **Wrong?** `pp_check()`: simulated vs. observed data.
3. **Accurate?** MAE, scaled MAE, interval coverage; use cross-validation / ELPD for honest, out-of-sample estimates.

Choose $k = 5$–10 (or LOO), and match folds to the data's dependence structure.

More practice questions ...

Question 1

A posterior predictive check shows that the simulated datasets match the observed data almost perfectly, and all MCMC diagnostics look excellent.

Can we conclude the model is good and ready to deploy? Why or why not?

Question 2

Model A has MAE $= 1000$ rides with predictive SD ≈ 1500. Model B has MAE $= 1000$ rides with predictive SD ≈ 300, and within_95 $= 0.60$.

Which model better describes its own uncertainty, and why is MAE alone not enough to judge them?

Question 3

For the bike model, in-sample MAE was 990 but 10-fold CV MAE was 1029.

Why did the error go up under cross-validation, and which number should we report when predicting future days?

Question 4

In a PPC for the bike model, about 3% of simulated days have *negative* ride counts. A colleague suggests running the MCMC chains 10 times longer.

Will that fix the problem? What does the PPC tell you, and what would you change?

Question 5

Two models fit to the same bike data:

- Model A (temperature): ELPD $= -4289$
- Model B (temperature + humidity): ELPD $= -4286$
- `loo_compare()`: difference $= 3$, SE of difference $= 4$

Is Model B clearly better at predicting new days? Which would you choose, and why?

Answers

Answer 1

No, not yet.

- Good MCMC diagnostics only say we approximated the *posterior* well; they say nothing about the model.
- A good PPC addresses *how wrong*, but not *how fair*: the data may exclude groups or encode bias, and the model's use may cause harm.
- It also says little about *out-of-sample* accuracy. We still need cross-validation / ELPD.

All three questions must be answered.

Answer 2

Model A.

- Same raw error, but Model B claims far more certainty than it has: its scaled errors are $\approx 1000/300 > 3$ SDs.
- within_95 $= 0.60 \ll 0.95 \Rightarrow$ B's intervals are too narrow (overconfident).
- MAE measures only the *center* of the prediction; scaled MAE and interval coverage evaluate the whole *posterior predictive distribution*.

Answer 3

- In-sample evaluation reuses the data the model was fit to, so the model is "graded on its own homework" $\Rightarrow$ optimistic.
- CV tests each fold on data the model never saw, mimicking prediction for genuinely new days.
- Report the **CV estimate (MAE $\approx$ 1029 rides)**; it is the more honest measure of future performance.

Answer 4

No. Longer chains improve the *MCMC approximation*, not the *model*.

- The posterior is already well approximated; a Normal model simply allows $Y < 0$.
- The PPC says the **data structure assumption is wrong**: ride counts are non-negative counts.
- Fix the model: Poisson / Negative Binomial regression (Ch. 12), or transform Y (e.g. $\log Y$).

MCMC quality and model quality are separate questions.

Answer 5

Not clearly.

- The difference (3) is smaller than its SE (4): well within noise, far from ≈ 2 SE.
- Judge the *difference* and its SE, not each model's ELPD and SE separately.
- The two models predict about equally well $\Rightarrow$ prefer the simpler Model A, unless humidity matters for another reason (interpretation, theory, a specific use case).